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EMPIRICAL STUDY

The Longitudinal Development of Fine-Phonetic Detail: Stop Production in a Domestic Immersion Program

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Abstract: This study explored the initial stages of adult second language (L2) learning with a special focus on the acquisition of the target language sound system. The aim was to analyze the longitudinal development of Spanish stop voicing contrasts in an immersion learning context. Native English-speaking late learners of Spanish provided production data for Spanish stops on a weekly basis throughout a domestic immersion program. The data were analyzed using Bayesian multilevel models and generalized additive mixed models. The analyses revealed phonetic learning in voiced stops over the course of the immersion program, though learners' production was inconsistent and did not fall within native ranges for most stop segments. The results suggest that L2 phonetic category formation can occur at an early stage of development and follows a nonlinear trajectory that is subject to individual and segment-specific variability. Furthermore, L2 phonetic representations are unstable during the early stages of learning.

Keywords sequential language learning; second language acquisition; longitudinal research; stop production; domestic immersion; phonetics

Introduction

Adults who learn a second language (L2) often have difficulties producing L2 phones (Flege, 2003; Flege & Hillenbrand, 1984; Flege & Wayland, 2019;

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Oyama, 1976). Current L2 speech models posit that these difficulties are explained in part by acoustic similarities or differences between the segments or contrasts of the first language (L1) and the L2 (i.e., Escudero & Boersma, 2004; Flege, 1995). For example, a native English speaker learning Spanish typically has difficulties with Spanish stops, which are phonemically similar but phonetically different from English stops. Importantly, at some point during the learning process, production begins to approximate that of the L2, though learners experience differing degrees of success (see Bongaerts, 1999). Current lines of research have attempted to explain how L2 phonetic categories develop over the early stages of L2 learning. The present work examined the beginning phase of adult L2 learning with a special focus on the production of voice timing in Spanish stops. Spanish contrasts phonetically voiced /b, d, g/ with phonetically voiceless /p, t, k/. English differs in that the phonetic implementation of the same contrasts occurs between lag stops (i.e., stops in which glottal pulsing begins after the release). This discrepancy in voice timing causes difficulties for L2 learners (Flege & Eefting, 1987a; Zampini, 2008, among many others). The present research was concerned with understanding when during the acquisition process adult learners begin to acquire new, language-specific phonetic categories and with understanding the amount of exposure necessary to develop this L2 fine-phonetic detail. Specifically, this work has contributed to knowledge of these issues by examining the beginning stages of adult learning throughout the course of a domestic immersion program. Domestic immersion provides an understudied linguistic context in which L1 use is minimized, L2 use is maximized, and L2 input is abundant. The experiment undertaken for this work has added to understanding of the initial stages of the acquisition of L2 phonology in adult learners.

Background Literature

Second Language Speech Models

Explaining native language influence on L2 production, perception, and spoken word recognition is at the heart of numerous theoretical models that account for the difficulties encountered by L2 learners during the acquisition of L2 phonology. Though there are many, for example, Best's perceptual assimilation model (Best, 1995; Best & Tyler, 2007) or the native language magnet model (Kuhl, 1992), the present work focuses on the speech learning model (Flege, 1995) and the L2 linguistic perception model (Escudero, 2009; Escudero & Boersma, 2004). These models are perceptually driven and maintain that the human ability to learn novel speech sounds remains active throughout the lifespan. Importantly, they have proposed that L2 difficulties are accounted for via phonetic similarities, differences, or perceived equivalences to native phones and contrasts, specifically, for these models, L2 learners' success in L2 production is grounded, at least initially, in their ability to discriminate L2 phones and contrasts.

The speech learning model posits that representations of the sounds of the new system being learned are stored in long-term memory and share a common phonological space with the L1. This fact implies that L1 to L2, as well as L2 to L1, interactions occur, though bilinguals' developing systems are thought to maintain L1 and L2 phonetic categories separately. Crucially, this model is perceptually driven. As the L1 sound system develops and perceptual attunement takes place, the formation of novel L2 phonetic categories becomes more difficult, though not impossible. One of the principal tenets of the model is that the human ability to learn new speech sounds is maintained throughout life. Phonetic similarity or perceived equivalence—what Flege called the equivalence classification—can predict L2 difficulties. In other words, the speech learning model is primarily concerned with (dis)similarities between L1 and L2 sounds. For instance, if, despite phonetic differences, a learner is unable to discriminate a L2 sound from a L1 sound—the two sounds are too perceptually similar—then category formation is blocked and production and/or perception of the segment is hypothesized to be difficult. On the other hand, if a learner can distinguish the relevant segments, the model proposes that a new phonetic category develops.

The L2 linguistic perception model, for its part, operates under the functional phonology framework (Boersma, 2001). At the core of functional phonology is the assumption that there are separate grammars for speech perception and speech production as well as distinct processing systems. The perception grammar is responsible for mapping characteristics of the acoustic signal to phonological representations. The L2 linguistic perception model maintains that, in the early stages of L2 learning, a new L2 grammar is created via full copying or full access, and then develops independently of the L1 perception grammar. Adjustments are made to the L2 perception grammar as the learner progresses in the language through a comparison module called the gradual learning algorithm. L2 phonological learning is explained via contrasts, rather than individual sounds. Concretely, there are three distinct learning situations. On the one hand, learners can perceive a contrast as being familiar—what Escudero refers to as a similar scenario, in which case they must then reset the boundary between the acoustic characteristics of this contrast via the gradual learning algorithm. The model hypothesizes similar contrasts are easier to learn because listeners can simply reuse L1 categories. Differences in fine phonetic detail between contrasts are accounted for through boundary adjustments. On the other hand, learners can also perceive a contrast as being novel—an instance of a new scenario. If this is the case, the L2 linguistic perception model contends that new categories must be formed, that is, L1 categories cannot simply be reused. The third possibility—the subset scenario—occurs when a L2 contrast is mapped to multiple L1 categories.

The speech learning model and the L2 linguistic perception model share common elements. For instance, both models make use of the acoustic signal—as opposed to articulatory gestures—regarding how segments or contrasts are assimilated. According to both models, L2 learners have the ability to become nativelike in their perception and production of the L2. Importantly, the two models also diverge in several ways. As mentioned, the L2 linguistic perception model posits full copying of the L1 perception grammar, which allows for L2 development to occur independent of the L1 perception grammar. This notion is at odds with the speech learning model, which proposes L1 and L2 categories share the same phonetic space, and therefore, predicts L1-L2 bidirectional phonetic interactions. In other words, L2 development is hypothesized to occur simultaneously with changes in L1 categories. The L2 linguistic perception model does not predict L1 perception or production to be affected by L2 phonological development. The speech learning model is used commonly in the L2 speech literature to explain the phonetic behavior of bilinguals and learners of all levels. The L2 linguistic perception model has mainly been used to model the acquisition of vowels and, according to Escudero (2005), is best utilized with longitudinal data collected from beginning language learners as they progress over time. For these reasons, both models were used to generate the research questions that guided the present work. Importantly, this research did not attempt to falsify or reconceptualize either model but rather drew on them to provide an account for the early stages of L2 phonological acquisition. Additionally, this work has extended the L2 linguistic perception model to consonants.

Longitudinal Development of Phonetic Categories

Recent work has explored how nonnative segments develop in L2 speech over time. Nagle (2017b) investigated the development of bilabial approximants in 26 English native-speaker L2 learners of Spanish. Spanish /b, d, g/ surface as approximants except after a pause or when they follow a homorganic consonant (i.e., a nasal or lateral preceding /d/). Importantly, the bilabial and velar approximants, [β, ɣ], are not part of the American English sound inventory, and the coronal segment has phonemic status. Thus, English-speaking L2 learners of

Spanish must develop new phonetic categories for the bilabial and velar approximant phones and learn that [ð] is an allophone in complementary distribution with [d]. Nagle's (2017b) study included university students matriculated in second or third semester Spanish classes and analyzed their production during five experimental sessions over an academic school year. Growth curve analysis was used to model the phonetic learning of [β] over time and showed the group trajectory to be flat, that is, as a group, the participating learners did not introduce approximants into their Spanish production. Despite the flat group trajectory, an analysis of individual differences showed nine of the 26 did produce [β] by the end of the investigation.

Other lines of longitudinal research have focused on voicing distinctions in stop contrasts. Cross-linguistically, stop contrasts vary primarily regarding voice onset time (VOT), though other acoustic cues differentiate stop identity, namely, closure duration (Raphael, 2005), F0 at the release (Kingston & Diehl, 1994), and the duration of the preceding vowel (Chen, 1970). The present work focused on VOT, which refers to the duration of the time interval between the release of the stop burst and the onset of voicing (Lisker & Abramson, 1964). For example, voicing can begin before the release of the burst (lead VOT), or it can begin after the release (lag VOT). Lag VOT can be sub-classified as a function of the duration of VOT. Stops with shorter VOT—which can range from 0 to 30 ms—are classified as short-lag and stops with long(er) VOT—typically 30 ms and longer—are classified as long-lag. These phonetic differences have important implications regarding speech production or perception because L2 learners produce or perceive target language stops as a function of the voicing characteristics of their L1 (Flege & Eefting, 1987b; Flege & Port, 1981). Moreover, accurate production of L2 stops can significantly improve foreign accent ratings by native-speaker judges (Sundara, Polka, & Baum, 2006). As target language productions deviate from native-speaker values, they are more likely to be characterized as nonnative (González-Bueno, 1997; Sundara et al., 2006).

Numerous studies have focused on the acquisition of voicing contrasts by analyzing VOT. Holliday (2015) explored 12 native Mandarin speakers' production of the Korean three-way stop contrast. On one hand, Mandarin has a two-way voicing contrast manifested through short-lag voiced and long-lag voiceless segments. On the other hand, Korean has a three-way voicing contrast in which VOT and F0 of the following vowel are critical cues. Holliday's (2015) investigation found that the majority of the participating learners did not use pitch in their production of Korean stops during the initial stages of learning. Six participants were tested a year later and three of them had acquired the

F0 cue. Holliday (2015) concluded that the Mandarin speakers' difficulties learning Korean stops was explained, at least partially, by the presence of multiple acoustic cues that are simultaneously at play. When the task facing the learner involves acquiring a new phonetic category consisting of a segment that is phonologically different though phonetically similar to a L1 sound, it appears to be rather difficult, even over an extended course of time.

Learning other L1-L2 pairs can involve segments with the same phonemic status. For instance, Spanish and English have a two-way voicing contrast. Concretely, these languages maintain a phonologically similar voicing distinction realized at different places of articulation: bilabial, coronal, and velar. A crucial difference between Spanish and English is how the stop voicing distinction is realized phonetically via VOT. English contrasts /b, d, g/ with /p, t, k/ through short-lag VOT and long-lag VOT, respectively. English /b, d, g/ can also surface as phonetically voiced segments, though this is less common and typically found only in prosodically weak positions and in connected speech (see Davidson, 2018). Spanish, on the other hand, contrasts /b, d, g/ with /p, t, k/ through lead VOT (or pre-voicing) and short-lag VOT, respectively (Lisker & Abramson, 1964). Thus, from the perspective of native English speaking L2 learners of Spanish, it is necessary to associate short-lag VOT with Spanish voiceless /p, t, k/ and conceivably create new phonetic categories for /b, d, g/ in order to produce or perceive the Spanish stops accurately.

In the L2 models described in the previous section, Spanish stops represent distinct challenges to native English speakers. Under the speech learning model, L2 segments that are acoustically similar to L1 categories are predicted to be difficult to learn. Though native English speakers are presumably familiar with phonetically voiced and voiceless segments, the speech learning model suggests that the Spanish stops will be difficult to the extent that they are perceived as being acoustically similar to the corresponding English categories. Under an L2 linguistic perception model account, the one-to-one phonological mapping suggests that Spanish stops represent a case of the easier, similar scenario, that is, L1 categories can be reused and the gradual learning algorithm will adjust perceptual boundaries as learners progress.

The literature has suggested that learners mitigate the challenges associated with acquiring new phonetic categories with varying degrees of success. Traditionally researchers have examined the acoustic properties of L2 speech and compared them to those of native, monolingual speakers. For example, Flege and Eefting (1987b) examined Spanish speakers' production of word initial English stops. The study found that the Spanish speakers produced English stops with aspiration. However, acoustic analyses showed that VOT values were longer than in Spanish words but not as long as VOT values of a monolingual English control group. Flege and Eefting (1987b) concluded the Spanish-English bilinguals did indeed develop L2 phonetic categories, but their productions were not "authentic" in that they still differed from those of the monolingual controls (p. 68). They ascribed these differences to the nature of the input their participants had received, specifically, Spanish-accented English. More recent research has attempted to understand how this process develops over time.

Nagle (2017a) examined bilabial stop production in the same 26 native English-speaking L2 learners of Spanish participating in Nagle (2017b). He found that these learners reduced VOT over time, suggesting that they may have developed language-specific phonetic categories for [b] and [p]. Though some participating learners had flat trajectories, the analyses also revealed that, as a group, learning trajectories followed a quadratic function, implying phonetic development occurred toward the beginning phases of testing and then slowed down over time. This finding corroborated previous research suggesting that L2 phonological acquisition may occur during the early stages of learning (see Chang, 2012; Holliday, 2015; Schuhmann & Huffman, 2015). Furthermore, Nagle (2017a) showed that, over the course of an academic year, the beginning stages of phonetic category formation can take place in a classroom learning context. This was noteworthy because in this context there is rarely explicit instruction regarding L1-L2 sound system discrepancies, overall L2 use is low, and learners receive minimal amounts of native input outside the classroom. How early phonetic category formation can occur in adults is still an open question.

The Role of Context

The past 30 years have seen a substantial increase in the number of students going abroad to learn a L2. Despite the relatively long history of study abroad, only in the past 25 years have researchers begun to seriously consider the benefits this type of learning has on L2 outcomes (Freed, 1995) and how these benefits directly compare to those offered in the traditional, at-home L2 classroom context. Overall, there is a growing body of evidence supporting the anecdotal suggestions of L2 teachers that the best way to learn a L2 is by doing so in situ (though see Nagle, Morales-Front, Moorman, & Sanz, 2016). Specifically, in the realms of oral proficiency (Collentine et al., 2004; Segalowitz & Freed, 2004), oral fluency (DeKeyser, 1986; Freed, Segalowitz, & Dewey, 2004), and cognitive abilities (Collentine, 2004; Lord, 2006) there has been a recognized benefit to studying abroad. However, there is also ample evidence pointing to an at-home advantage for some areas such as grammatical structure and morphosyntactic control (see Collentine & Freed, 2004; DeKeyser, 1986). There have been mixed results with regard to lexical development (Collentine, 2004; DeKeyser, 1986) and pragmatic and communicative abilities (Collentine et al., 2004; Lafford, 2004). There is, nonetheless, a clear gap in the literature regarding studies on L2 phonology (though see Sardegna, Lee, & Kusey, 2018), presumably due to the fact that pronunciation has taken a back seat in a large portion of the L2 acquisition literature (Bajuniemi, 2013; Porter, 1999).

The majority of the investigations examining learning context and L2 phonological acquisition have focused on consonants (Bajuniemi, 2013; Díaz-Campos, 2004, 2006; Lord, 2010; Stevens, 2001; Zampini, 1998, among others), though there is a growing body of research on vowels (Avello & Lara, 2014; Simões, 1996; Stevens, 2011). For instance, Díaz-Campos (2004) analyzed late learners' production of a series of Spanish segments as a function of learning context (at-home vs. study abroad). The study found that the groups did not differ in their production of voiceless stops, but rather improved equally. Neither group showed significant improvement for the voiced approximants [β, ð, ɣ]. A follow-up analysis suggested that the study abroad group showed more targetlike pronunciation in informal, conversational speech compared with that in more formal registers (Díaz-Campos, 2006). Other studies have shown production can also improve in at-home contexts when teaching is complemented with explicit pronunciation instruction (Bajuniemi, 2013), but much less so without it (Zampini, 1994). In a similar vein, stop production was shown to improve significantly in the study abroad context when learners took a phonetics course prior to traveling abroad (Lord, 2010), but this does not appear to be obligatory for improvement to be observed (cf. Stevens, 2001). Other investigations on laterals and palatal nasals have shown substantial production improvements in both at-home and study abroad contexts (Díaz-Campos, 2004).

Research looking specifically at the acquisition of Spanish has also focused on the at-home context (González López & Counselman, 2013; López, 2012; Nagle, 2017a; Zampini, 1998; Zampini & Green, 2001), the study abroad context (Martinsen & Alvord, 2012; Simões, 1996), or directly compared the two (Díaz-Campos, 2004, 2006; Martinsen, Alvord, & Tanner, 2014; Martinsen, Baker, Dewey, Bown, & Johnson, 2010; Stevens, 2011). Work on voice timing of stops has been particularly fruitful. For instance, Zampini (1998) used a semi-longitudinal design to study the production and perception of Spanish and English bilabial stops in intermediate and advanced late learners over the course of a semester in the at-home context. Her research showed that her participants produced Spanish /b/ with a shorter VOT than for the English counterpart but with a longer VOT than did native Spanish speakers (see also Zampini & Green, 2001, for other cues). By the end of the semester the participants had still not incorporated pre-voicing into their production of /b/. Nonetheless, the results of Zampini (1998) supported the notion that phonetic category formation is possible in the at-home context.

Stevens (2001) found similar results in the study abroad context. This study explored the production of Spanish /p, t, k/ by a group of North American English speakers who participated in a study abroad program in Spain. Stevens (2001) compared their production of Spanish stops at the end of the program to that of an at-home control group. The study found that the study abroad group learned to reduce aspiration and that the at-home group continued producing the stops with long-lag VOT. A more recent investigation by Nagle et al. (2016) also looked at Spanish stops in a pretest-posttest design. The learning context was a 6-week intensive immersion program. The study found that 18 advanced learners reduced VOT for all stop segments, suggesting that, in this context, phonetic category development can take place over a relatively short period of time in advanced learners.

Another learning context that has received less attention has been domestic immersion. Domestic immersion programs differ from traditional study abroad in that they take place in learners' country of residence. There has been a dearth of research examining this context, and the studies that have been conducted have focused on the development of listening, reading, and writing skills, along with general fluency (see Collentine et al., 2004; Dewey, 2002; Liskin-Gasparro, 1998; McKee, 1983). To the best of my knowledge, there are no studies examining the acquisition of L2 phonology. Domestic immersion likely includes many of the same advantages of study abroad, with the benefit of being more accessible to people for whom it may not be feasible to spend time abroad (i.e., professionals, parents, etc.). One such program, Middlebury College's Language Schools, takes place over the course of 7 weeks during the summer. The program offers intensive immersion classes that attract motivated students at all levels in a number of different languages from around the world. Of particular interest is the fact that enrolled students must sign a formal agreement—the Language Pledge—by which they promise to use only the target language for the duration of the program. Students take the pledge seriously and use the target language exclusively. It is often cited as the primary reason why students choose to attend Middlebury's Language Schools.[1] If students fail to comply with this language pledge, they can be expelled from the program. The classes at Middlebury use a communicative language focus with instruction entirely in the language being studied. Students are exposed to numerous varieties of the language that they are studying but are typically taught a standard variety in the classroom. Outside of class, students participate in co-curricular activities, for example, theater, sports, dance, in which they use the target language in authentic communicative settings. In sum, this domestic immersion program provides an input-rich environment sought out by highly motivated students and represents an understudied learning context in which L1 use is prohibited and, importantly, L2 use and input are maximized. The present study was set in this context.

The Present Study

The goal of the present study was to investigate the acquisition of fine phonetic detail over the course of the beginning stages of L2 phonological development. Specifically, this research considered the following questions.

1. Is there evidence that beginning adult L2 learners can acquire new phonetic categories in a 7-week domestic immersion program? Previous research has suggested that phonetic category development can occur in the at-home context (see López, 2012; Zampini & Green, 2001, among others), and in the study abroad context (see Díaz-Campos, 2004; Lord, 2010; Stevens, 2001, among others), though most studies have dealt with intermediate and advanced learners (cf. Martinsen et al., 2014). Thus, this research aimed to fill a gap in the current knowledge of the early stages of L2 phonological development by studying absolute beginners.
2. In light of evidence of phonetic category development, how much exposure is necessary for the acquisition of the L2 fine-phonetic detail associated with stop voicing? Previous studies focusing on intermediate and advanced learners have demonstrated pronunciation gains over the course of an academic semester (i.e., Stevens, 2001; Zampini, 1998) and an academic school year (i.e., Nagle, 2017a). It remains to be seen how much L2 pronunciation can develop during the early stages of learning in optimal conditions.
3. Finally, what is the nature of the initial learning trajectories of phonetic category development? Current research on learning trajectories for stop segments has suggested that intermediate learners' development unfolds as a quadratic function of time (i.e., Nagle, 2017a).

Little is known about the time course of development for Spanish stop voicing in absolute beginners. To this end, the present work analyzed the ongoing acquisition of Spanish stops in native English-speaking adult L2 learners of Spanish. First, this research analyzed how the production of the stop voicing contrasts changed over time. Next, the trajectories of phonetic learning for voiced and voiceless stops were scrutinized, and, finally, the learners' stop production was compared directly to a control group of native bilinguals in order to shed light on how nativelike Spanish stops were produced after intense exposure. The adult learners participated in the Middlebury 7-week domestic immersion program in which L1 use was prohibited. They completed a production task in a longitudinal research design in order to evaluate the early phonetic category development of the Spanish stop voicing contrasts in an understudied learning context in which L1 use was minimal and L2 input and use was maximal.

Method

Questionnaires

Two questionnaires were administered to the learners: a language background questionnaire and a weekly progress questionnaire. The background questionnaire served to exclude participants not meeting the recruitment criteria, namely, individuals who had already studied Spanish or other languages. The weekly progress questionnaire provided self-report data about the participants' use of Spanish and English, speaking abilities, listening and comprehension abilities, overall abilities in Spanish, whether or not they felt their Spanish had improved, satisfaction with progress, and whether they participated in co-curricular activities. It was administered every Sunday at the beginning of the experimental session. Participants' responses were in reference to the previous week. The Bilingual Language Profile (Birdsong, Gertken, & Amengual, 2012), was administered to a bilingual control group. The Bilingual Language Profile measures language dominance by calculating scores for four modules: language history, language use, language proficiency, and language attitudes. The scores from each module are summed and provide an estimate of language dominance ranging from −218 to 218, with the end points indicating dominance in one of the two languages. For this study, negative scores were associated with dominance in English, and positive scores were associated with dominance in Spanish. Scores near the mid-point (0) were taken as an indication of balanced bilingualism.

Participants

The present study included production data from 20 participants who were divided into two groups. The first group consisted of 10 adult native speakers of English who were learning Spanish as a L2 in the Middlebury domestic immersion context. The second group, a control group, comprised 10 simultaneous Spanish-English bilinguals.

Late Learners

Based on information from the background questionnaires, 10 learners (six female) who reported (a) minimal prior experience with other languages (i.e., no more than a semester) and (b) not having spent a significant amount of time in a foreign country (i.e., longer than a short vacation) were selected. The learner group comprised functionally monolingual young adults ($M_{\text{age}} = 23.70$ years, $SD = 5.27$) with limited exposure to Spanish or any other languages. They were evaluated as absolute beginners based on a placement test and oral interviews with two faculty members of Middlebury's Language Schools program. The participants' instructors were monolingual speakers of peninsular Spanish (see Appendix S1 for additional information).

Figure 1 displays a radar plot summarizing the assessment data over the course of the program. This adult learner group used their L1, English, minimally, and their L2, Spanish, almost exclusively. The input to which the learner group was exposed was provided mainly by native speakers or nonnative speakers with higher levels of proficiency in Spanish. The assessment questionnaire also showed that these adult L2 learners believed their listening, speaking, and overall abilities in Spanish improved over the course of the program.

Simultaneous (Native) Bilinguals

The control group comprised 10 simultaneous (i.e., native) bilinguals[2] that reported being exposed to both English and Spanish from birth and speaking both languages since they had learned to talk. Their parents were first generation immigrants and also bilingual. The native bilingual participants stated that they used both languages daily with friends and family. Their ability to speak Spanish was confirmed via interactions with the author before testing. The mean language dominance score was 7.80 (95% CI [−3.64, 18.78]), suggesting rather balanced bilingualism. The mean proficiency score was 0.45 (95% CI [−3.40, 4.31]), suggesting that they did not believe they were more proficient in one language over the other. Finally, the mean language use score was 1.60 (95% CI [−9.81, 11.01]), corroborating that they used both languages with similar frequency and had comparable Spanish-English competency. Additional information is available in Appendix S1 of the Supporting Information.

Materials

The materials used in this study are available at https://osf.io/w5hsx/.

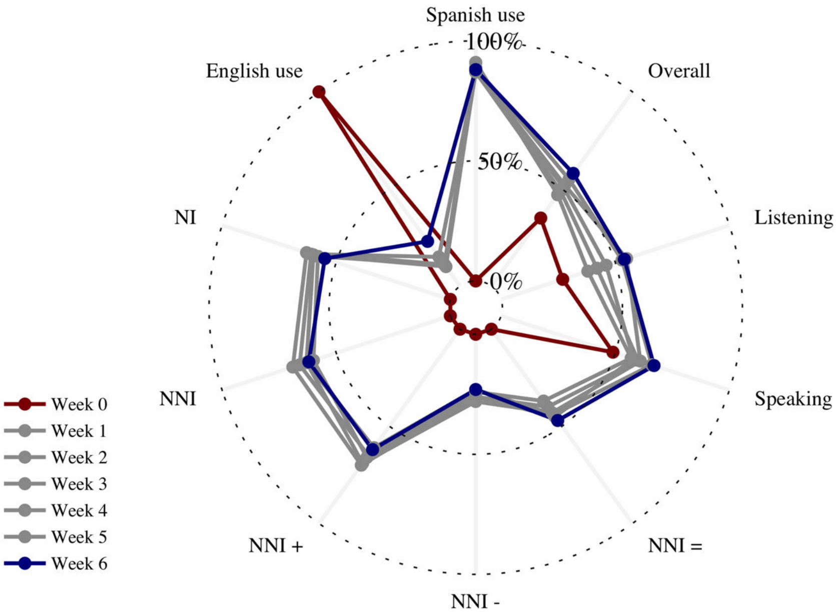

Figure 1 Radar plot of the self-assessment questionnaire data. Each line represents a week in the program. Each point represents a mean score, scaled to range from 0 to 1, for a given metric of the questionnaire. The metrics include: mean Spanish/English use, estimated native input (NI), estimated nonnative input (NNI), estimated NNI from speakers with a higher level (NNI+), estimated NNI from speakers with a lower level (NNI−), estimated NNI from speakers with the same level (NNI =), and estimated speaking, listening, and overall ability (in Spanish). [Color figure can be viewed at wileyonlinelibrary.com]

Target Phrases

The experiment included 30 nonce words containing stop segments in word initial position. The target words were embedded in the carrier phrase "____ es la palabra" (____ is the word). There were five targets for each stop. All targets had CV.CV syllable structure and paroxytonic stress. The stops were followed by one of the five Spanish vowels /i, e, a, o, u/ and then the consonant /k/. The final vowel was always the same as the previous vowel (i.e., /ˈba.ka/, /ˈbe.ke/, /ˈbi.ki/, etc.). There were six stops for each of the five vowels, yielding a total of 30 items. The target words were interspersed among 20 distractors: 12 real words used in another task and eight pseudowords words with a V.CV syllable structure.

Auditory Stimuli

The auditory stimuli used in the delayed repetition portion of the experiment were provided by a 29-year-old female native Spanish speaker from Cádiz, Spain. The speaker was a Spanish-dominant simultaneous (native) bilingual (Spanish-English) visiting the United States from Spain when the stimuli were recorded. A Shure SM10A dynamic head-mounted microphone was used to record the items. A Sound Devices MM-1 pre-amplifier boosted the signal and sent it to a laptop computer where it was recorded using Praat (Boersma & Weenink, 2018) at a 44.1 kHz sample rate with 16-bit quantization. The recording took place in a sound attenuated booth in a phonetics laboratory.

Recordings

The production data elicited from the learners were recorded in a quiet classroom at Middlebury College. A Shure SM10A dynamic head-mounted microphone was used to record the participants' productions. A Sound Devices MM-1 pre-amplifier passed the signal to a Marantz PMD661 MKII handheld solid state broadcast recorder. The recordings were sampled at 44.1 kHz with 16-bit quantization. The same setup captured the native bilingual group's speech in a sound attenuated booth.

Acoustic Analysis

The participants' productions were segmented in Praat following standard procedures. Synchronized waveform and spectrographic displays were used to hand mark the onset of voicing, the burst for each of the stops, and the onset and offset of the following vowel. The onset of voicing was taken to be the first periodic pattern found in the waveform. The criterion for bursts was the onset of broadband sudden noise in the spectrogram. A Praat script automatically extracted VOT by calculating the difference (in ms) between the onset of voicing and the burst. The same script automatically calculated relative VOT by dividing the aforementioned VOT value by the duration (in ms) of the CV (stop + vowel) sequence.[3]

Procedure

The participants completed four distinct experiments, one of which was a production experiment, the focus of the present study. The initial experimental session took place on the second or third day of the immersion program. During the initial session, the participants completed the first iteration of the assessment questionnaire, a delayed repetition production task, and a two-alternative forced-choice perception task (not reported). From that point forward until the final week of the program, the participants completed the same tasks with the exception that the delayed repetition production task was replaced with a reading production task. Two separate tasks were used in this experiment: delayed repetition during week 0 and reading during the remaining weeks. Delayed repetition was used in week 0 in order to avoid intervocalic *t* being realized as [ɾ], which is standard in American English, in the V.CV distractors. These distractors served as target stimuli for another experiment. The sessions took place every Sunday with the exception of the last week, which included multiple sessions and two separate experiments not reported here. The final session took place on a Wednesday, which was 3 days after the seventh session and 3 days before the program ended.

The native bilingual participants were recruited from a university in the southwestern United States. Their participation included 2 days of testing with a minimum of 24 hours between sessions. On the first day, they completed the Bilingual Language Profile questionnaire, the two-alternative forced-choice task (not reported here), and the reading task. On the second day, they completed the tasks not reported here.

PsychoPy2 (Peirce et al., 2019) presented the experimental stimuli randomly. The first session employed a delayed shadowing technique, the stimuli being presented aurally via the audio recordings of the native Spanish speaker. The participants listened and repeated the 50 items, which were embedded in the aforementioned carrier phrase, in three randomized blocks. Subsequent experimental sessions used a reading task. In these cases, the stimuli appeared on the screen of a laptop computer, and the participants read them aloud. After saying the stimuli aloud, the participants pressed a button on a keypad to advance to the next item. Each participant provided the dataset with 720 utterance initial stops (six stops × five vowels × three repetitions × eight experimental sessions). Thus, a total of 7,200 stop tokens were collected from the participants (720 tokens × 10 participants = 7,200). The native bilingual group completed a single session of the reading task, and provided a total of 900 utterance initial stops (six stops × five vowels × three repetitions × 10 participants). All participants finished the task in approximately 10 minutes.

Results

The results are divided into three subsections dealing with (a) phonetic development over time, (b) learning trajectories, and (c) comparisons of the adult learner participants with native bilingual participants. An analysis of individual differences is provided in Appendix S3 of the Supporting Information. Given the distinct nature of the analyses used in each subsection, a brief description of the statistical approach has been provided at the beginning. All analyses were conducted in R (Version 3.6.0; R Core Team, 2018; session information is available in Appendix S4).

The Time-Course of Phonetic Category Development

The first analysis examined stop production over the course of the immersion program in order to determine if or when phonetic learning took place. The production time-course data were analyzed using Bayesian multilevel models fitted in Stan via the R package brms (Bürkner, 2017).[4] VOT was modeled as a function of exposure time in days (0, 7, 14, 21, 28, 35, 42). Exposure time was dummy coded, and day 0 was set as the reference level. The random effects structure included by-subject and by-item intercepts with by-subject random slopes for exposure time and by-item random slopes for exposure time and item repetition. The same model formula fit the data for each stop segment with 2,000 iterations (1,000 warm-up). Hamiltonian Monte-Carlo sampling was carried out with four chains distributed between 16 processing cores in order to draw samples from the posterior predictive distribution. The models included regularizing, weakly informative priors (Gelman, Simpson, & Betancourt, 2017). Specifically, all parameters were assumed to be distributed as normal with a standard deviation of 50. Standard deviation parameters for random effects and model residual error (sigma) were truncated to exclude negative values. The correlation parameter used a LKJ correlation prior with regularization set at 2. The results for voiced stops are presented first followed by the voiceless stops.

Voiced Stops

Figure 2 plots the voiced stop data as a function of exposure time and place of articulation. The plots show a decrease in the mean values that began around 21 days of exposure. From that point forward, more pre-voicing—seen as negative VOT values—was present in the participants' production of all three segments. Importantly, the presence of short-lag stops (low, positive VOT values) appeared concurrently with the pre-voiced stops. In other words, the raw data suggested variable production of voiced segments, consistent with voiced segments in Spanish and English, beginning after approximately 21 days of exposure.

Figure 3 provides the plots of the VOT point estimates of the posterior distributions as a function of exposure time for /b, d, g/, along with the 70% and 95% credible intervals. The point estimates for all three segments were negative after 21 days of exposure and from that point onward: /b/, $\beta = -28.42$,

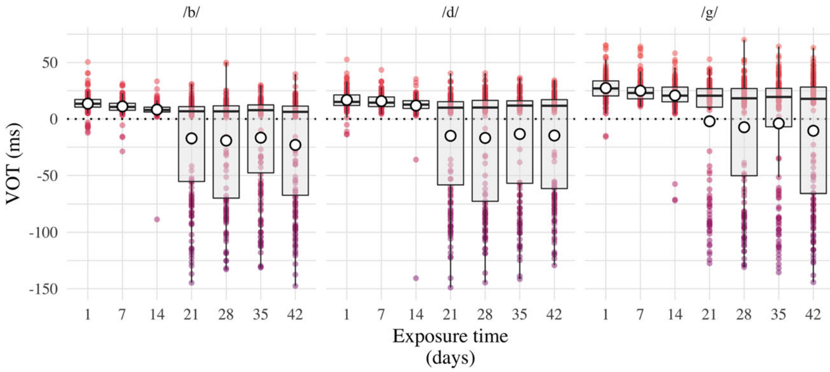

Figure 2 Boxplots of raw voice onset time (VOT) as a function of exposure time for voiced stops. White circles represent mean values. [Color figure can be viewed at wileyonlinelibrary.com]

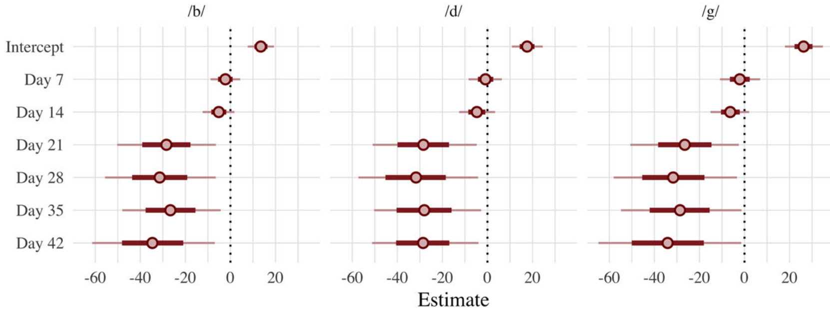

Figure 3 Posterior point estimates ±70% and 95% credible intervals for voiced stops as a function of exposure time. The intercepts represent the day 1 baseline values. The parameter estimates for days 7–42 represent differences in ms from baseline values. [Color figure can be viewed at wileyonlinelibrary.com]

95% CI [−50.26, −6.44]; /d/, $\beta = -28.43$, 95% CI [−50.96, −4.79]; /g/, $\beta = -26.52$, 95% CI [−50.68, −2.50]. The 95% credible intervals were wide and in all three cases, no longer contained 0, giving strong evidence to support the notion that VOT values reduced as a function of exposure time. However, inspection of the raw data also showed that production was rather unstable, varying between negative values and low positive values. Table 3, available in Appendix S2, provides the point estimates and 95% credible intervals for all experimental sessions across the three segments.

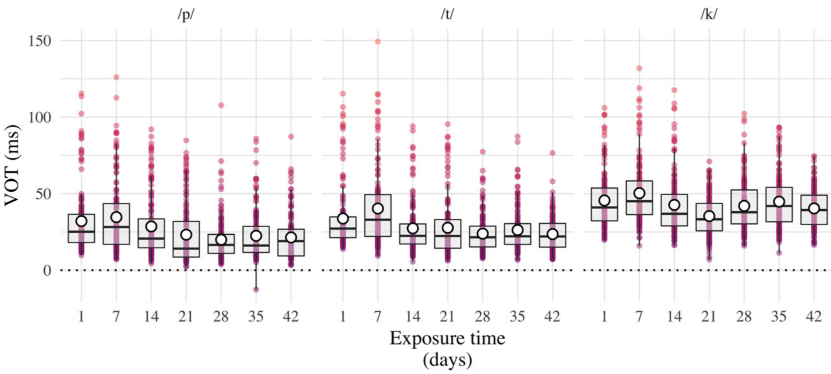

Figure 4 Boxplots of raw VOT as a function of exposure time for voiceless stops. White circles represent mean values. [Color figure can be viewed at wileyonlinelibrary.com]

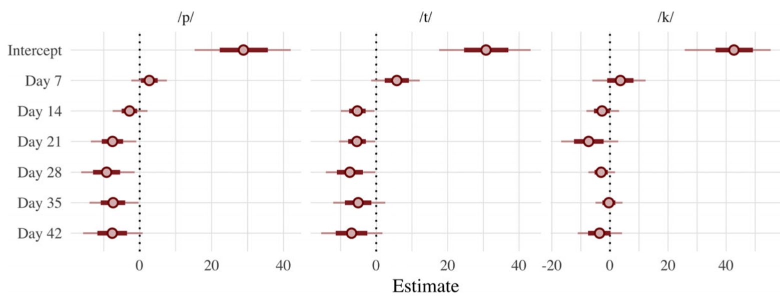

Figure 5 Posterior point estimates ±70% and 95% credible intervals for voiceless stops as a function of exposure time. The intercepts represent the day 1 baseline values. The estimates for days 7–42 represent differences from baseline values. [Color figure can be viewed at wileyonlinelibrary.com]

Voiceless Stops

Figure 4 provides the plots for the voiceless stop data as a function of exposure time and place of articulation. In this case, the plots depict a more tenuous overall decrease in the mean VOT values. There was no clear point after which VOT was clearly different from the baseline values. In fact, after initial exposure, mean values seemed to increase slightly (see day 7). Overall, the variability of the voiceless productions appeared to reduce as a function of exposure time, as indicated by the reduced spread of the distribution of raw VOT values after 28, 35, and 42 days of exposure.

Figure 5 displays the plots for the VOT point estimates of the posterior distributions as a function of exposure time for /p, t, k/, along with the 70%

and 95% credible intervals. The point estimates for all three segments tended to be slightly negative, though there was high variability and all of the credible intervals included 0 at the end of the immersion program. In other words, there was weak evidence supporting the notion that these late learners, as a group, reduced aspiration in voiceless stops after 42 days of exposure. The segment that was produced with the greatest decrease in VOT was /p/ (approximately 9.21 ms lower after 28 days of exposure). Table 4 of Appendix S2 provides the point estimates and 95% credible intervals for all experimental sessions across the three segments.

Together, the analyses of the time-course data suggested that these late learners' production of voiced stops became more nativelike as exposure to Spanish increased (i.e., they incorporated pre-voicing into their production of /b, d, g/. Specifically, their phonetic behavior was clearly different after 21 days of exposure. By the end of the program more improvement was seen in the voiced bilabial, followed by the coronal, and then the velar. For voiceless stop production, on the other hand, VOT decreased minimally, suggesting that these late learners still aspirated /p, t, k/ at the end of the immersion program, with the voiceless bilabial showing most improvement.

Learning Trajectories

To shed light on the nature of the phonetic development trajectories over the course of the immersion program, the production time-course data were analyzed using generalized additive mixed models (Sóskuthy, 2017; Winter & Wieling, 2016, see Appendix S2 for more information). For the learning trajectory analysis, the VOT time-course data were standardized and modeled as a function of the parametric term voicing (voiced, voiceless) and a nonlinear function of exposure time. Voicing was set as an ordered variable with voiced stops coded as 0. Cubic regression splines with four basis knots were applied (a) as a reference smooth to exposure time, (b) as a difference smooth to exposure time conditioned on voicing, and (c) as a random smooth for each participant conditioned on exposure time. This specification used the trajectory of the voiced stops as the baseline and compared it to the trajectory of the voiceless stops. Given that the VOT values were standardized as a function of voicing, there was no voicing effect on the intercept (both voiced and voiceless stops have an overall mean of 0 and a standard deviation of 1), though the model was informative regarding how the shape of the voiced and voiceless learning trajectories differed from each other as a function of exposure time. All analyses were conducted in R using the mgcv package (Wood, 2004) for model fitting and itsadug for visualization (van Rij, Wieling, Baayen, & van Rijn, 2017).

Table 1 Summary of generalized additive mixed models output reporting parametric coefficients (Part A) and the effective degrees of freedom and reference degrees of freedom, F and p values for smooth terms (Part B)

A. Parametric coefficients	b	SE	t	p
Intercept	0.01	0.14	0.07	>.05
Intercept Δ voiced	0.00	0.02	−0.09	>.05
B. Approximate significance of smooth terms	**EDF**	**Ref. DF**	***F***	***p***
Reference smooth: Exposure time	1.78	1.85	4.59	<.01
Difference smooth: Voiced	2.75	2.94	20.91	<.01
Random smooth: Exposure time × subject	34.76	38.00	59.49	<.01

Note. The full model fit voice onset time as a function of voicing. $R^2 = .32$; deviance explained = 32.89%; n = 6,283. EDF = effective degrees of freedom; Ref. DF = reference degrees of freedom.

Autocorrelation was inspected visually using autocorrelation function plots of model residuals. Hypothesis testing for effects of voicing were conducted using a combination of t tests and approximate F tests on parametric and smooth terms, respectively, in conjunction with nested model comparisons.

The voicing model explained 32% of the variance and 32.89% of the deviance. Nested model comparisons suggested that the parametric and smooth terms on modality significantly improved fit ($df = 3$, difference = 28.15, effective $df = 9, p < .001$; see Table 5 of Appendix S2 for the complete output of the nested model comparison). Table 1 provides the complete model output. Part B of Table 1 shows that the voicing trajectories varied as a function of exposure time. The value higher than 1 for the effective degrees of freedom (EDF) indicated that the trajectory was nonlinear. Crucially, Part B further shows that the voiced stop trajectory differed from the voiceless stop trajectory, the EDF value indicating that the difference between the trajectories was also nonlinear.

Figure 6 provides the plots for the voicing trajectories (left side) and the estimated differences between them over the time course (right side). The plots corroborate the findings derived from the model. Specifically, both stops followed a nonlinear time course. The voiceless stops shifted earlier and at a consistent rate before the slope flattened out around 25 days of exposure. The voiced stop boundaries, on the other hand, followed a more sigmoid-like trajectory, flattening out slightly after the voiceless trajectory. The estimated difference plot extrapolates the numeric time predictor over 100 values ranging from 1 to 42 and indicates two exposure time windows of significant differences:

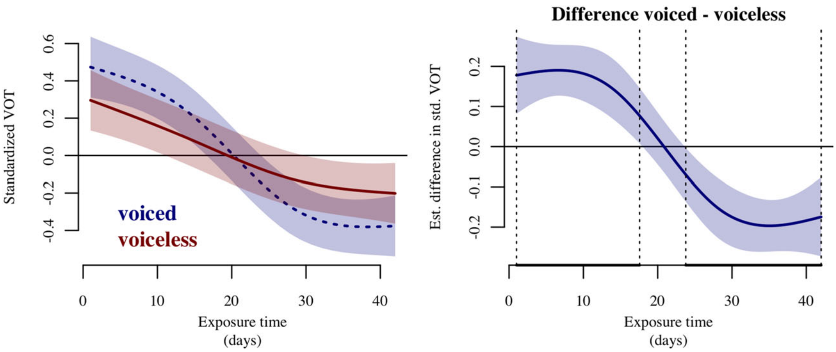

Figure 6 Nonlinear trajectories of voiced and voiceless stops (left panel) and the estimated difference (voiced − voiceless) as a function of exposure time. The dashed line represents voiced stops (left panel). VOT = voice onset time. [Color figure can be viewed at wileyonlinelibrary.com]

from days 1 to 18 and days 24 to 42. These time windows are highlighted in the plot by the vertical, discontinuous lines. Overall, the analysis suggested that phonetic category development began at an early stage during these adult participants' L2 learning and that voiced and voiceless stop segments followed different learning trajectories for them.

Comparison With Native Bilinguals

The learner data from the final experimental session were compared directly to the native bilingual control group using Bayesian multilevel models fitted in Stan via the R package brms (Bürkner, 2017). Relative VOT was the criterion variable and was modeled as a function of group (learner, native bilingual) and standardized speech rate.[5] The group predictor was sum coded (−1, 1); thus, the posterior distribution of parameter estimates provided a sampling distribution of plausible values for the difference between groups in standard units at the mean speech rate. The random effects structure included by-subject and by-item intercepts with by-subject random slopes for speech rate and by-item random slopes for item repetitions. The same model formula fit the data for each stop segment with 2,000 iterations (of which 1,000 warm-up). Hamiltonian Monte-Carlo sampling was carried out with four chains distributed between 16 processing cores in order to draw samples from the posterior predictive distribution. The models also included regularizing, weakly informative priors (Gelman et al., 2017). All parameters were assumed to be distributed as normal with a standard deviation of 10. The prior for model residual error (sigma) was

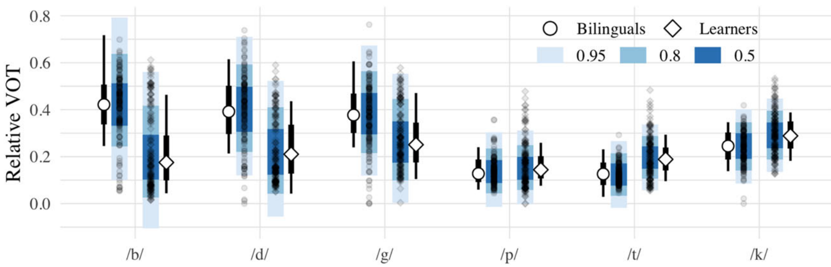

Figure 7 Relative VOT as a function of group and segment. The plot includes posterior point medians ±66% and 95% quantile credible intervals (white points + black line); along with 95%, 80%, and 50% posterior predictive intervals (blue); and the raw data. [Color figure can be viewed at wileyonlinelibrary.com]

modeled using Student's *t* distribution with mean of 2 and standard deviation of 0.1.

Figure 7 shows the plots for the raw voiced and voiceless stop data as a function of group (learner, native bilingual) and segment (/b, d, g, p, t, k/), along with descriptives of the corresponding posterior distributions. There was a clear pattern based on voicing. On the one hand, native bilingual VOT for voiced stops occupied a larger portion of the CV sequence than did the VOT for the late learners' productions, indicating the bilingual controls had larger and negative lead VOT values. On the other hand, the late learners' VOT for voiceless stops tended to occupy a larger portion of the CV sequence compared to the portion occupied by the CV sequence of the native bilinguals' productions—particularly for /t/ and /k/, indicating the late learners had higher lag VOT values. This suggested that the late learners produced the voiced and voiceless stops differently from the native bilinguals. These observations were supported by the analysis. Specifically, the late learners pre-voiced /b, d, g/ less and aspirated /p, t, k/ more. The point estimates for β group were negative for all voiced segments and none of the credible intervals included 0, suggesting that the late learners' production of voiced stops differed from the native bilinguals across the board: /b/, $\beta = -0.22$, 95% CI [−0.34, −0.11]; /d/, $\beta = -0.19$, 95% CI [−0.30, −0.08]; /g/, $\beta = -0.12$, 95% CI [−0.21, −0.03]. For the voiceless segments the point estimates for β group were all positive with credible intervals that did not contain 0, with the exception of /p/. In other words, there was evidence that /t/ and /k/ were more aspirated in the L2 speech, but /p/ was just within range of the native bilinguals: /p/, $\beta = 0.02$, 95% CI [−0.01, 0.04]; /t/, $\beta = 0.07$, 95% CI [0.03, 0.11]; /k/,

$\beta = 0.05$, 95% CI [0.01, 0.09]. The complete table including parameter estimates for speech rate is available in Table 6 of Appendix S2. In sum, the analyses comparing the late learner participants' production of stop segments to that of the native bilingual participants showed that, after 42 days of exposure, the late learner participants' production generally still differed from that of the native bilinguals for all segments except the voiceless bilabial.

Discussion

Summary of Findings

The goal of the present work was to explore the early stages of L2 phonological acquisition in a domestic immersion learning context. The study included longitudinal VOT data for word-initial Spanish stops. Comparing the data from each experimental session to the baseline—day 0 initial state, the first analysis revealed that, overall, VOT decreased as exposure to Spanish increased. This was particularly true for the voiced segments /b, d, g/ and to a lesser extent for the voiceless bilabial, /p/. Crucially, the analysis suggested robust change in VOT after 21 days of exposure time, which typically continued until the end of the program. The distribution of the raw data suggested that the learners' production of /b, d, g/ was variable. After 21 days of exposure to Spanish, the participants began to incorporate pre-voicing in to their realizations of voiced stops, but not categorically, that is, there were still ample short-lag stop productions in the data. The voiceless stops, on the other hand, did not show a large decrease in aspiration, though the spread of the data did decrease over time, with the majority of the distribution falling more tightly around a lower center of gravity by the end of the program.

The second analysis focused on the learning trajectories for voiced and voiceless stops and provided evidence for three main findings: (a) for stops, learning trajectories were nonlinear functions of time and (b) stop segments followed different trajectories according to voicing. Specifically, voiceless segments followed a more shallow, quadratic function, but voiced stops followed a more sigmoid-like, cubic function, though there were individual differences in learning patterns. Taken together, these assertions suggest that (c) phonetic category development can occur at an early stage of L2 learning. This finding is complemented by the results of the first analysis of the time-course data in which robust production differences were observed in the voiced stops after only 21 days of exposure to Spanish.

The third analysis directly compared the late learner participants' production of Spanish stops at the end of the program to a group of native bilinguals. Overall, the late learners clearly differed from the native bilinguals for nearly all segments. Importantly, the voiced stops included many productions that fell within the native range. Of the voiceless stops, the late learners' production of /p/ was the only segment that lay (just) within the range of the native bilingual productions.

A series of analyses of individual differences (see Appendix S3) showed that there was within-group variability in L2 stop voicing development. This was particularly true for the time course of stop learning, where development unfolded as nonlinear functions of time that varied in functional form. Furthermore, the analyses suggested that individuals who reported receiving more native input over the course of the domestic immersion program also produced stops with relative VOT that was closer to that of the native bilinguals after 42 days of exposure to Spanish (see Figure 13 of Appendix S3).

Interpretation and Implications

The present work focused on responding to three research questions. Regarding the first question, "Is there evidence that beginning adult L2 learners can acquire new phonetic categories in a 7-week domestic immersion program?," the present work clearly provides evidence of changes in voice timing of Spanish stops that is consistent with phonetic category development. In response to the second question, "In light of evidence of phonetic category development, how much exposure is necessary for the acquisition of the L2 fine-phonetic detail associated with stop voicing?," the analyses suggest that voice-timing began to differ from baseline values after 21 days of exposure. Finally, the present work asked a third question, "What is the nature of the initial learning trajectories of phonetic category development?" The longitudinal data suggest that phonetic category development of Spanish stops unfolds as a nonlinear function of time that differs depending on voicing, though the functional form of the trajectories is susceptible to individual differences.

These findings suggest that phonological acquisition can take place in adult learners of Spanish in a domestic immersion context. The results draw parallels with those of Nagle (2017a), where adult learners reduced VOT for bilabial stops over the course of an academic year. A crucial difference between the two studies is that in the present work evidence of phonetic learning was obtained over a much shorter period of time. This may be explained by the fact that (a) the participants in the present study received more L2 input and (b) used their L1 minimally and the L2 maximally. In the context of the present work, minimal L1 use equated to approximately 14% of the time based on participant self-report (see Figure 1). In Nagle (2017a), the phonetic learning trajectories were quadratic in nature for both voiced and voiceless segments. In the present study only the voiceless stop trajectory followed a quadratic function. Nagle (2017a) included five time points over the course of an academic year, whereas the present investigation included seven time points over the course of 42 days. Thus, temporal resolution may explain the voicing difference in learning trajectories. That is to say, both studies observed similar effects; however, having more experimental sessions over a shorter period of time allowed the present study, in a sense, to zoom in on the learning process. Another crucial difference between the two studies is that Nagle's (2017a) study did not include absolute beginners, in their initial state. Thus, the voicing differences observed here may be a part of early phonological learning that occurs when L2 input and use are maximized and L1 use is minimized.

Research related to L2 outcomes in distinct learning contexts has shown a clear advantage for study abroad with regard to oral proficiency, oral fluency, and cognitive abilities. Conversely, the traditional classroom setting has proven advantageous regarding grammatical structure and morphosyntactic control. The present work contributes to the literature examining the acquisition of L2 phonology in a domestic immersion learning context. Research conducted by Zampini (1998) suggested that phonetic category development can occur in the traditional classroom when explicit articulatory instruction is provided. Research by Stevens (2001) has shown that gains similar to those observed in Zampini (1998) are attainable in the study abroad context in the absence of explicit pronunciation instruction. The results of the present work suggest that the domestic immersion context appears to offer the same benefits for stop production as those found in the foreign immersion context. Both foreign and domestic immersion contexts provide the learner with high levels of input. It is possible the nature of the input available differs; however, depending on numerous factors related to the personal preferences of each participant. For instance, in an study abroad context, a learner conceivably could be immersed with native input, but for any number of reasons choose to use the L2 minimally. Thus, it may be the case that L2 learning is accelerated in a domestic immersion context where L2 use is an obligatory part of the program, at least with regard to phonological acquisition.

It is important to note that after 7 weeks of speaking Spanish exclusively, the late learners in the present investigation did not perform in a native bilingual-like fashion for most of the stop segments, nor were they expected to do so. Decades of research related to ultimate attainment in adult L2 learners have suggested that nativelike production is uncommon. In the cases in which adult learners have achieved levels of production or perception indistinguishable from that of native speakers, they also reported high amounts of L2 input and low L1 use (i.e., Flege, MacKay, & Meador, 1999). Other research has shown that individuals who are highly skilled language learners also show high levels of motivation (Bongaerts, 1999). With this in mind, it seems feasible to hypothesize that highly motivated adult learners, to whose profile the Middlebury students can be assumed to correspond, would have become increasingly more nativelike were the immersion program longer. It is unclear how much more time would be necessary, or how much more improvement could be made. Both the speech learning model and the L2 linguistic perception model suggest that learning will continue. However, in a strict sense, the speech learning model predicts that nativelike attainment is unlikely because, under this model, L2 phones are hypothesized to share the same phonetic space with L1 phones; thus, continued cross-language interference (both L1 > L2 and L2 > L1) is expected. Moreover, acquiring a new phonetic category is believed to be difficult as L1 categories are established in long-term memory. For this reason another point of interest revolves around what occurs upon leaving the immersion context. Are L2 gains maintained, or do L2 phonetic categories deteriorate with limited use?

For voiced stops, the late learner participants showed a large rate of change in VOT but were still far from the bilingual native average. Importantly, for all of the late learners, many of the voiced stops did fall within the native range, suggesting that they did learn to incorporate pre-voicing into their speech, but did so inconsistently, likely due to cross-linguistic interference (and/or different contexts of production). In other words, at this early stage, L2 phonetic representations are still unstable. The voiceless stops, on the other hand, can only decrease to a certain extent in the timeframes examined here. The analyses showed that the late learners' /t, k/ still had longer relative VOTs than those of the native bilingual participants.

What can explain this voicing difference? Under a speech learning model account, one possibility is that the voiced stops of Spanish are easier to learn for English speakers than voiceless stops. This could be explained by the difference between creating a new phonetic category versus readjusting or reassigning an already established category. In other words, equating already familiar short-lag realizations with voiceless stop categories might be more challenging than equating unfamiliar pre-voiced realizations with voiced categories if learners perceive the phones (pre-voiced and short-lag) as being similar. This hypothesis, however, is based on the notion that pre-voiced stops are unfamiliar to English speakers. Though they are less common, phonetically voiced realizations of /b, d, g/ can in fact occur in English in prosodically weak positions and in connected speech (Davidson, 2018). Under a L2 linguistic perception model account, an individual's tendency to pre-voice may predict a different learning task because L2 segments would map to a single L1 voiced category. In other words, the acoustic to phonetic form connection would present a new scenario for voiced stops in which phonetically voiced and voiceless stops both activate L1 voiced categories in the initial state. If an individual does not pre-voice in this position in their L1, then the L2 linguistic perception model would predict an easier learning task, namely, a similar scenario. The present research did not include English production data; thus, no definitive claim can be made concerning whether the late learners pre-voiced in their L1 in such contexts. Such information may provide insight into explaining individual differences in future studies.

The L2 linguistic perception model contends that nativelike abilities are possible for all learners and that the L1 will not be affected by L2 learning. This is because the L1 categories are believed to be copied and used for L2 production and perception in the beginning stages of learning. Therefore, they are kept separate and subsequently adjusted to ambient L2 input with exposure. Research by Chang (2013, 2019) has provided evidence of L1 phonetic drift in beginner and advanced learners of Korean during initial stages of learning and 52 weeks after an immersion program. Future research could extend these findings to beginning learners of Spanish by including formal analyses of the learners' English, both before the immersion program and after, in order to offer evidence to support or refute the hypothesis that L2 learning affects native categories. The early gains observed in the learner group mesh well with the copy-and-adjust nature of the L2 linguistic perception model account, especially given that decreases in VOT for voiceless stops do occur eventually (i.e., Nagle, 2017a). However, more longitudinal data showing L1 categories are affected by L2 learning (e.g., Chang, 2019) are necessary to provide evidence that the L1 and L2 perceptual systems are not the separate systems that are proposed by the L2 linguistic perception model. This, in turn, would lend support to the speech learning model's account of longitudinal phonetic category development, whereby the L1 system can be affected by L2 learning.

Limitations and Future Research

The present study followed 10 adult learners over the course of an intensive immersion program. Longitudinal data such as these offer many benefits to the L2 speech literature. However, the advantages come at a cost. In particular, sample size and participant retention are issues (see Ortega & Iberri-Shea, 2005). Future research would benefit from including more participants to ensure and promote replicability (Bergmann et al., 2018; Roettger, Winter, & Baayen, 2019) when examining the early stages of L2 phonological learning.

The present work did not include any formal analyses of the learners' L1 English, neither before the immersion program nor after it. Future research could include measures of English stop production in unambiguous English-sessions at the start and conclusion of the immersion program. This would provide further evidence for phonetic category development and allow for a clearer understanding of cross-linguistic interference during the beginning stages of learning and over time.

Another point of interest involves the speech segments under investigation. This work focused on the development of stop voicing in languages that include the same phonemic stop contrasts. Future longitudinal investigations ought to extend the previously mentioned findings to other segments, such as laterals, nasals, and vowels in order to derive a more complete picture of the learning development trajectories of L2 segments. Of particular interest for English L1 speakers learning Spanish are the longitudinal development of spirantization of the voiced coronal, /d/, and the acquisition of the rhotic trill, /r/. The former represents a phonemic contrast in American English (as opposed to allophonic variation in Spanish), and the latter is not part of the American English phonemic inventory. Thus, these segments provide examples in which category realignment and category formation, respectively, would be necessary for production and perception, and neither instance is likely to be explained via boundary resetting. In essence, the acquisition of these segments could provide an interesting point of comparison to stop voicing, and the amount of exposure necessary for category formation to occur could shed light on the relative difficulty of boundary resetting versus learning a new allophonic distribution versus learning a new sound altogether.

Finally, it would be beneficial to analyze late learners' production of Spanish stops weeks, months, and years after their program ends in order to better assess the rapid gains observed in domestic immersion and compare them to those observed in a study abroad context. Relatedly, the present study analyzed a learning context in which L1 use was low, L2 use was high, and L2 input was omnipresent. Research is underway that is comparing a domestic immersion learning context directly with study abroad and the traditional classroom using longitudinal designs. In so doing, this line of research will contribute to understanding the effects of L1 and L2 use and the relative importance of different types of target language input on the development of L2 phonology.

Conclusion

The present work investigated the initial stages of adult L2 learning with a special focus on the acquisition of the target language sound system. Specifically, this study analyzed the longitudinal development of the Spanish stop voicing contrasts in a learning context in which L1 use was minimal and L2 use and input were maximal. The experiment undertaken for this work adds to knowledge of the acquisition of L2 phonology in adult learners, and, more specifically, to understanding the ongoing development of the fine-phonetic detail of Spanish stop voicing. The data from the study indicate that L2 phonetic representations are particularly unstable during the early stages of learning. Furthermore, the longitudinal data suggest that L2 phonetic category formation can occur at an early stage of development and follow a nonlinear trajectory that is subject to individual and segment-specific variability.

Final revised version accepted 16 November 2019

Notes

1 More information regarding the pledge is available at http://www.middlebury.edu/ls/academics/language-pledge

2 Traditionally L2 acquisition research has used monolingual speakers as comparison groups for L2 learners in speech production or perception. The use of bilingual control groups is part of a recent trend in L2 phonetic research (see Sakai, 2018) that attempts to provide a fairer and more useful assessment of learner progress by comparing their abilities to those of a target population (bilinguals) that represents their learning goals (bilingualism).

3 Relative VOT represents the proportion that VOT occupies in the duration of the stop-vowel sequence. Relative VOT has proven to be a more accurate, fine-grained metric for comparing stop voicing between groups because it controls for possible speech rate differences (see Stölten, Abrahamsson, & Hyltenstam, 2015). Relative VOT is measured as a proportion of the stop-vowel sequence or as a proportion of the target word. For Spanish voiced stops, one expects higher relative VOT values because they are pre-voiced. Voiceless stops, on the other hand, should have lower relative values because the short-lag targets occupy a smaller proportion of the CV sequence.

4 Bayesian data analysis has emerged as an alternative to frequentist data analysis (see Vasishth & Nicenboim, 2016). The details behind how Bayesian data analysis works are beyond the scope of the present study. The interested reader is encouraged to consult Kruschke and Liddell (2018), Nicenboim and Vasishth (2016); Norouzian, de Miranda & Plonsky (2018) for descriptions and tutorials designed for linguists, as well as van de Schoot and Depaoli (2014) for a general presentation of reporting results for the social and behavioral sciences.

5 Speech rate can affect voice timing of stops (Kessinger & Blumstein, 1997) such that faster speech typically results in reduced VOT. To control for possible confounds of speech rate in group comparisons, speech rate was measured as the average number of syllables produced over the duration of a given utterance (De Jong & Wempe, 2009). There was no evidence for a difference in speech rate between these late learner and native bilingual participants, $\beta = 0.08$, 95% CI [−0.42, 0.38] (see Figure 9). Nonetheless, standardized speech rate was entered into the model as a covariate so that parameter estimates for possible group differences in relative VOT would be calculated at the overall mean speech rate ($M = 1.44$ ms, $SD = 0.66$).

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This article has earned Open data and Open Materials badges for making publicly available the components of the research methods needed to reproduce the reported procedure. All materials that the authors have used and have the right to share are available at https://osf.io/w5hsx/. All proprietary materials have been precisely identified in the manuscript.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1. Additional Information Regarding Middlebury College and VOT Across Varieties of Spanish
Appendix S2. Additional Statistical Information
Appendix S3. Analyses of Individual Differences
Appendix S4. Session Information for Reproducibility

Appendix: Accessible Summary (also publicly available at https://oasis-database.org)

Adults Can Show Improvement in Pronunciation of Spanish After a Short-Term Domestic Immersion Program

What This Research Was About and Why It Is Important

This research explores how adults learn to pronounce the sounds of a second language (L2). Numerous studies suggest learning an L2 as an adult is difficult and having a "foreign accent" is practically inevitable. However, we know little about how much adults can improve when they receive large amounts of L2 input and focus on using the L2 exclusively, i.e., when they do not speak their first, native language. This study explored how much adults' pronunciation of Spanish improved when participating in a presumably ideal learning context: immersion. Specifically, this research tracked the progress of 10 adults in a domestic immersion program—one that took place in the US—in which they were not allowed to use English. That is, they studied Spanish, participated in extra-curricular activities in Spanish, and spoke exclusively in Spanish, nonstop, for 7 weeks. This research is important because it sheds light on how adults learn L2 pronunciation in optimal conditions and the roles played by L2 input and use. By the end of the program the 10 adults improved their pronunciation of Spanish, though it still differed from a native control group, and some sounds improved more than others.

What the Researcher Did

- The learners were 10 adults, native English speakers from the US participating in a 7-week domestic immersion program in Spanish. They had no proficiency in Spanish prior to beginning the program.
- The learners studied Spanish in the mornings. During the afternoon and evening, they participated in activities such as dance, soccer, theater, etc., in Spanish. The learners lived in the residence halls of Middlebury College (Middlebury, VT, USA) with other students and professors.
- The use of English, or any language other than Spanish, was strictly prohibited.
- The researchers investigated the Spanish sounds known as "stops": /b, d, g/ (voiced) and /p, t, k/ (voiceless). These have complex similarities and differences with the pronunciation of similar sounds in English.
- The learners' pronunciation was tested on a weekly basis throughout the domestic immersion program.
    - They produced real and Spanish-like words and were recorded using professional audio equipment.
    - Each week their pronunciation was compared to their baseline, day 0 pronunciation, and to a control group of native Spanish speakers in order to track their progress.

What the Researcher Found

- Overall, the learners' pronunciation was better at the end of the program.
- Most participants began to show improvement after 21 days in the program.
- Some sounds were learned fast and "all at once" (the voiced stops), while others showed slower, more linear progress (the voiceless stops).

- Though they did improve, at the end of the program, the learners' pronunciation was inconsistent and still differed from the native Spanish speakers.

Things to Consider

- This research considers an understudied learning context (domestic immersion) which allows the researcher to better control L2 use and input. Future investigations should compare contexts, such as the traditional classroom and study abroad immersion, in order to shed more light on how L2 use and input influence how adults learn to pronounce the sounds of other languages.
- Longitudinal research in second language acquisition, particularly during the early stages of learning, provide a nuanced look at how pronunciation in an L2 develops. L2 pronunciation research would benefit from more longitudinal approaches with larger sample sizes in order to further our understanding of the development of different L2 sounds and the challenges they pose.

Materials and data: Available at https://osf.io/w5hsx/

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